

Evaluation of Serum Magnesium, Calcium, Phosphate and Antioxidant Vitamins in Sickle Cell Patients in Kano State-Nigeria

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ABSTRACT

Sickle cell disease (SCD) is characterized by frequent electrolyte imbalances, which can exacerbate disease complications, particularly during vaso-occlusive crises (VOC). These imbalances, coupled with oxidative stress due to reduced antioxidant vitamins, contribute significantly to SCD patho-physiology. This study aimed to investigate the serum levels of key electrolytes—calcium, magnesium, and phosphorus—as well as antioxidant vitamins A, C, and E in SCD patients during crises, those in steady state and healthy controls in Kano, Nigeria. A comparative analysis was conducted on serum samples from SCD patients in crisis 30(37.5%), those in steady state 30(37.5%), and healthy controls 20 (25%). Serum calcium, magnesium, and phosphorus levels were measured spectrophotometrically, along with antioxidant vitamins. Statistical comparisons were made using ANOVA, and Pearson correlation was employed to analyze relationships between electrolyte levels. The SCD patients in crisis and steady state exhibited significantly lower serum calcium ($p=0.035$) and magnesium ($p=0.004$) levels, and higher phosphate ($p=0.016$) compared to healthy controls. The study also found significantly reduced levels of antioxidant vitamins A, C, and E in SCD patients, particularly during crises ($p<0.05$). Correlation analysis revealed strong positive relationships between electrolyte levels, suggesting parallel physiological changes in SCD. The findings highlighted significant disruptions in electrolyte and antioxidant vitamin levels during VOC in SCD patients. These imbalances may contribute to the disease's severity and progression. Dietary antioxidants may offer therapeutic benefits in restoring red-ox balance and mitigating oxidative stress in SCD.

Keywords: Sickle cell disease, electrolytes, Antioxidant vitamin

INTRODUCTION

Sickle cell disease (SCD) is a severe genetic disorder, particularly prevalent in sub-Saharan Africa, caused by a mutation in the β -globin gene leading to abnormal haemoglobin S (HbS). This results in red blood cells assuming a sickle shape under low oxygen conditions, causing vessel blockages. It leads to complications such as vaso-occlusive crises, chronic haemolytic anaemia, and tissue damage, posing a significant public health challenge.¹

The clinical spectrum of sickle cell disease includes several forms: Sickle Cell

Anemia (HbSS), Sickle Haemoglobin C Disease (HbSC), and Haemoglobin S Beta-Thalassemia Disease (HbS β Thal). These conditions manifest when an individual inherits sickle cell traits from both parents, leading to the production of mutant haemoglobin that disrupts normal blood flow and causes premature destruction of red blood cells.²

In addition to the direct effects of sickling, SCD is associated with a variety of systemic complications. These include bone disorders such as osteopenia and osteoporosis, which are associated with increased risk of fractures, vertebral collapse and bone pain⁴. The underlying pathophysiology involves increased erythropoiesis, hypoxia due to vessel obstruction, and consequent tissue injury, which adversely affects bone health.³

Electrolyte imbalances and disturbances in bone metabolism are frequently observed in SCD patients. Reports from

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various studies indicate anomalies in serum levels of calcium, magnesium, and phosphate. For example, there is one study that documented hypocalcaemia and hyperphosphatemia in Nigerian children with SCD, while in another study, low magnesium levels and a high calcium-to-magnesium ratio in adults with SCD patients has been observed.^{5,6,7} Other research has highlighted the persistent deficiency of magnesium and calcium, which are crucial for maintaining bone integrity and overall health.^{8,9,10}

Magnesium plays a critical role in numerous physiological processes including bone health. It contributes to electrolyte balance, muscle function, energy metabolism, and cell division¹¹. Magnesium deficiency has been associated with a range of bone disorders, including osteoporosis, due to its effects on bone remodeling and calcium homeostasis.^{11,12} Similarly, calcium and phosphorus are essential for skeletal integrity and various physiological functions such as muscle contraction and neuronal excitation. Disruptions in the homeostasis of these minerals can lead to significant bone-related issues.¹³

In the context of sickle cell disease, disruptions in calcium and phosphorus levels, combined with the effects of repeated vaso-occlusive crises and bone infarcts, contribute to the development of conditions such as avascular necrosis and degenerative arthritis.¹⁴ The oxidative stress associated with SCD further complicates the disease. Sickle cell patients experience increased oxidative stress due to heightened production of reactive oxygen species (ROS) and reduced antioxidant defences. This imbalance leads to oxidative damage, which exacerbates haemolysis and contributes to disease complications.^{15,16} Antioxidant vitamins such as vitamins A, C and E play a crucial role in neutralizing ROS and protecting cells from oxidative damage¹⁵. Studies have shown that SCD patients have decreased levels of these antioxidants, which further emphasizes the need for monitoring and potentially supplementing antioxidant vitamins as part of the disease management strategy.¹⁷

Nigeria, with its high prevalence of sickle cell disease, presents a critical context for this research. With an estimated 24% prevalence of sickle cell trait and 100,000 new SCD births annually, understanding and addressing the nutritional and metabolic imbalances in these patients is vital.¹⁸ Evaluating the serum levels of magnesium, calcium, phosphate, and antioxidant vitamins can provide valuable insights into the management of SCD and help in mitigating some of the disease's severe complications.

Therefore, sickle cell disease represents a complex interplay of genetic, metabolic, and oxidative factors that significantly impact patient health. This research will focus on the serum concentrations of key electrolytes (magnesium, calcium, and phosphate) and antioxidant vitamins (Vitamin A, C and E) in sickle cell patients in Kano State, aiming to better understand their role in disease progression and management. By examining these parameters, the study seeks to contribute to improved clinical strategies for managing sickle cell disease and enhancing patient outcomes.

MATERIAL AND METHODS

Equipment

The equipment that was used include; spectrophotometer (UV-2100 PC), micropipette, gel activator vacutainer tube, centrifuge (80-1), test-tubes, plain tubes.

Study area

The study was conducted in the outpatient clinics of Murtala Muhammed Specialist Hospital and Aminu Kano Teaching Hospital, both tertiary healthcare facilities in Kano State, Nigeria. Kano State, located in the Northwest geopolitical region, covers 20,760 square kilometers and has a population of 9.4 million, according to the 2006 census.^{19,20,21} The state borders Katsina, Jigawa, Bauchi, and Kaduna.

Ethical Consideration

Ethical clearance to conduct the research was obtained from the Kano State

Ministry of Health. The participant consent was obtained prior to the administration of questionnaires. All study was carried out in accordance with the declaration of Helsinki.²²

Study Population

The study participants were divided into three groups comprising sickle cell patient with crisis (group 1) and sickle cell patient without crisis (group 2) and healthy control subjects (group 3) both attending Murtala Muhammed Specialist Hospital, and Aminu Kano Teaching Hospital.

Bio-data and Informed Consent

Socio-demographic data and medical history were collected using structured interviews. Participants at the sickle cell clinic were informed about the study, and consent was obtained verbally and in writing.

Inclusion Criteria

Tested and confirmed Sickle cell patients with crisis (clinically on pain, fever, vasculitis, multi organ infections and low Hb with high TWBCs) who consent to participate. Sickle cell patients without crisis who consented to participate. Healthy aged matched control subjects

Exclusion Criteria

The study excluded SCD patients with chronic renal failure, hyper-parathyroidism and stroke

Research Design

The study is a hospital-based prospective comparative and cross-sectional which lasted for four months (January to April, 2024).

Sample Size Determination

The sample size was obtained using the formula for the calculation of minimum sample size used in a study involving human subjects/patients, which is:

$$n = \frac{z^2 pq}{d^2} \quad 23 \quad p = \text{prevalence of sickle cell disease} = 5.00\% = 0.05. \quad 24$$

Total samples = 80 samples,

Test = 60 samples (Patients with crisis = 30 samples, Patients without crisis = 30 samples)

Control = 20 samples.

Sample Collection, Preservation and Processing

About 5ml of venous blood sample was collected aseptically from the ante-cubital fossa of each participant using sterile syringe with the vacutainer tube containing gel and clot activator (gel tube). The blood samples were allowed to clot and then centrifuged at 350rpm for 5minutes. The separated serum was then transferred into pre labeled Serum container and stored at -20°C until ready for assay.

Biochemical Analysis of the Samples

Measurement of Serum Calcium

This was achieved by colorimetric method.²⁵

Principle: In alkaline solution, o-cresolphthalein complexone (CPC) forms a purple complex with calcium. Interference with magnesium ions is reduced by the addition of 8-hydroxyquinoline. The colour intensity of the complex formed is measured and is directly proportional to the calcium concentration in the sample.²⁵

propanol (Buffer), O-cresolphthalein complexone and 8-hydroxyquinoline.²⁵

Measurement of Serum Magnesium: This was achieved by Metallochromic dye Xylidyl Blue was used.²⁶

Principle: Serum magnesium ions react with Xylidyl Blue in an alkaline solution to produce a red complex that is measured spectrophotometrically. The intensity of color produced is directly proportional to magnesium concentration. Calcium interference is virtually eliminated by use of EGTA and a surfactant system is included to remove protein interference.²⁶

Reagent: When combined the reagent contains: xylidyl blue 0.1mM, EGTA 0.13mM, DMSO 1.4M, Buffer, surfactant, non-reactive stabilizers including potassium cyanide at 0.02% w/v. ²⁶

Measurement of Serum Phosphate

Serum phosphate was measured using colorimetric method.²⁷

Principle: The measurement of inorganic phosphorus in serum is usually accomplished

by forming a phosphomolybdate complex and in turn reducing it to a molybdenum blue color complex.²⁷

Reagent: Sulfuric acid Ferrous ammonium sulfate Ferrous nitrate Ammonium, Sulphuric acid, Ammonium molybdate and Phosphorous standard.²⁷

Measurement of Serum Vitamin A

Serum vitamin A was estimated by Spectrophotometrically.²⁸

Measurement of Serum Vitamin C

Serum vitamin C was estimated Spectrophotometrically.²⁸

Estimation of Serum Vitamin E

Serum vitamin E concentration was estimated Spectrophotometrically.²⁸

Statistical Analysis

The data were stored in a Microsoft excel 2016 and then analyzed using statistical package for the social sciences (SPSS) for windows version 23.0. The data were expressed as Mean $\pm$ Standard deviation (SD). The statistical significance between means were estimated using ANOVA when appropriate. Pearson's correlation coefficient (r) was used to measure the strength of the association between the variables. The $p < 0.05$ was considered to be significant.

RESULT

Sociodemographic Characteristics of the Participants

The socio-demographics characteristics of the participant shows that slightly above average of the participant were females (51.2%) and male constitute 48.8%, majority were of Hausa tribe (78.8%), with a significant portion falling within the age range of less than 10 years (43.8%). Additionally, a considerable proportion of participants had the SS genotype (75.0%), AS genotype (11.3%), 2 with AC genotype (2.5%), and 9 with AA genotype (11.3%), while the distribution across crisis statuses showed 37.5% of patients were on crisis, 37.5% patient on steady health, and

25.0% as health controls as at when the samples were collected as shown in table 1.

Comparison of the Biochemical Parameters of SCD Patients with Crisis and SCD Patient on Steady Health, as well as Healthy Control Subjects

ANOVA analysis revealed significant variations in serum electrolyte and antioxidant levels among sickle cell patients in crisis, those in steady health, and healthy controls. Sickle cell patients in crisis had notably lower serum calcium levels (1.20 ± 0.30 mmol/L) compared to steady-state patients (1.97 ± 0.16 mmol/L) and healthy controls (2.17 ± 0.18 mmol/L) ($p = 0.035$). Additionally, serum phosphate levels were significantly elevated during crises (2.19 ± 1.50 mmol/L) compared to steady health and controls ($p = 0.016$), while serum magnesium levels were markedly reduced (0.48 ± 0.21 mmol/L) during crises. Antioxidant vitamins A and E were significantly lower in sickle cell patients, with consistent Vitamin C levels in crisis and steady-state groups, but significantly decreased in comparison to controls.

Relationship Between Different Electrolytes in Sickle Cell Patients Disease Patients

The Pearson correlation analysis revealed varying relationships between electrolyte levels among two groups of SCD patient. There is strong positive between serum calcium and phosphate ($r=0.670$, $p=0.038$), calcium and magnesium ($r=1.000$, $p=0.024$), and phosphate and magnesium ($r=1.000$, $p=0.023$) in both groups of SCD patients as shown in table 3.

Antioxidant Vitamins among the Age groups of the SCD Patients

Table 4 illustrates the variations in serum antioxidant vitamins A, C, and E across different age groups among the SCD subjects, categorized as follows: 0-10, 11-20, 21-30, and 31 and above years. For SCD patients, the mean serum levels were as follows: Vitamin A was 0.84 ± 0.09 μ mol, 0.73 ± 0.09 μ mol, 0.68 ± 0.06 μ mol, and 0.68 ± 0.77 μ mol; Vitamin C was 21.88 ± 1.27 μ mol, 21.08 ± 1.58 μ mol,

Table 1: Socio-demographics Characteristics of the Participants

Variable	Frequency	Percentage (%)
GENDER		
Male	39	48.8
Female	41	51.2
AGE GROUP		
Less than 10years	35	43.8
11-20years	11	13.8
21-30years	27	33.8
31-40years	5	6.3
41-50years	1	1.3
51-60years	1	1.3
TRIBE		
Hausa	63	78.8
Fulani	7	8.8
Yoruba	4	5.0
Others	6	7.5
CRISIS STATUS		
Patient on crisis	30	37.5
Patient on steady health	30	37.5
Health control	20	25.0
GENOTYPE		
SS	60	75.0
AS	9	11.3
AC	2	2.5
AA	9	11.3
TOTAL	80	100

Table 2: Mean±SD of the Biochemical Parameters in study participants

Parameters	SCPC	SCPS	HC	P-value
Calcium (mmol/L)	1.20± 0.30 ^a	1.97± 0.16 ^b	2.17± 0.18 ^c	0.035*
Phosphate (mmol/L)	2.19± 1.50 ^a	1.77± 0.90 ^b	1.25± 0.44 ^c	0.016*
Magnesium (mmol/L)	0.48± 0.21 ^a	0.85± 0.22 ^b	1.24± 0.62 ^c	0.004*
Vitamin A (µmol/L)	0.53±0.19 ^a	0.77± 0.16 ^b	1.45± 0.04 ^c	0.011*
Vitamin C (µmol/L)	23.60±0.30 ^a	24.90± 0.35 ^a	32.98± 0.78 ^b	0.031*
Vitamin E (µmol/L)	9.41±0.65 ^a	13.49±0.28 ^b	20.22± 0.84 ^c	0.021*

Legend: SCPC, Sickle cell patients on crisis; SCPS, Sickle cell patients on steady health; HC, Healthy control, *-significant at $p < 0.05$

Post-hoc comparisons using the Tukey HSD test indicates that values with different superscript are significantly different (Y^a and Y^b), whereas values with the same superscript are not significantly different (Y^a and Y^a).

20.65±1.04 µmol, and 19.10±2.13 µmol; and Vitamin E was 10.87±0.62 µmol, 10.45±0.61 µmol, 9.92±0.30 µmol, and 9.90±10.52 µmol, respectively. Statistical analysis revealed no significant differences in vitamin levels across age groups, with p -values > 0.05 .

DISCUSSION

Electrolyte imbalances are prevalent in sickle cell disease (SCD) and pose significant risks to patient health. Essential electrolytes like calcium, magnesium, and phosphorus, which play vital roles in physiological functions, often show abnormal levels in SCD patients. These disturbances can exacerbate the complications associated with the disease, particularly during a sickle cell crisis—a condition marked by acute pain and tissue ischemia due to the blockage of small blood vessels by sickled red blood cells.²⁹

The current study investigated serum calcium, magnesium, and phosphorus levels as well as antioxidants vitamins (Vitamin A, C, and E) in SCD patients during crises, those in steady states, and healthy controls. The results revealed notable differences in these

electrolytes across the groups. Specifically, during a sickle cell crisis, patients exhibited lower serum calcium levels, while magnesium and phosphorus levels were elevated compared to both steady-state SCD patients and healthy individuals.

In SCD, calcium disturbances are closely linked to bone disorders such as osteoporosis and osteopenia, which are more common due to chronic hemolysis, inflammation, and impaired bone remodeling.^{5,6} The study found that serum calcium levels were significantly lower during crises compared to non-crisis patients and healthy controls. This decrease in calcium levels may result from an inflammatory response during crises, leading to increased calcium utilization by immune cells, dehydration, and reduced calcium absorption. Hypocalcemia have been frequently observed in individuals with sickle cell anemia (SCA), particularly during vaso-occlusive crises (VOC). Research indicates that hypocalcemia may be associated with reduced dietary intake in children with SCA, potentially contributing to the increased frequency of VOC.³⁰ Hypocalcemia during VOC in SCA children

prolongs QTc interval, increasing risk of ventricular arrhythmias and cardiac arrest.

Table 3. Relationship Between Different Electrolytes in Sickle Cell Disease Subjects

Variables	r- value	p-value
Patient on crisis		
Calcium and Phosphate	0.670	0.038*
Calcium and Magnesium	1.000	0.024*
Phosphate and Magnesium	1.000	0.023*
Patient on steady health		
Calcium and Phosphate	0.670	0.038*
Calcium and Magnesium	1.000	0.024*
Phosphate and Magnesium	1.000	0.023*

Legend: r- value, Pearson correlation; p-value, Probability value, *-significant at $p < 0.05$

(31,32). The findings of this study are consistent with previous research,^{6,9,33} which also reported lower serum calcium levels in SCD patients compared to control.

In this study we found low serum magnesium levels in SCD patients on crisis and steady-health compared to healthy controls. Reports from previous studies on the level of magnesium in sickle cell patients have been variable, ranging from low to normal.^{5,6,10} Some studies have shown that, magnesium levels are low in sickle cell anemia^{5,10} while others reported normal levels.⁶

Despite normal serum magnesium levels, decreased RBC magnesium concentrations have been documented in individuals with sickle cell disease (SCD), potentially due to increased activity of the Na/Mg exchanges in magnesium-loaded sickle erythrocytes.³⁴ Magnesium plays a crucial role in SCD beyond its effect on RBC dehydration, potentially influencing vascular tone through nitric oxide-mediated vasodilation and calcium inhibition in smooth muscle. Additionally, magnesium's ability to prolong clotting time might contribute to a reduction in vaso-occlusion.³⁵

Phosphorus plays a key role in cellular energy metabolism, bone mineralization, and nucleic acid synthesis. The study observed elevated serum phosphorus levels during crises compared to steady-state patients and healthy controls. Elevated phosphorus levels observed during crises may result from increased cell turnover and hemolysis, leading to the release of intracellular phosphate into the bloodstream. Additionally, the heightened metabolic demands of the inflammatory response during a crisis could enhance phosphate mobilization to support energy metabolism and cellular repair.¹³

This study is in consistent with one study,³⁸ that reported significantly higher serum phosphate in the sickle SCD patients compared to control. They related their findings to increased tubular reabsorption of phosphate in these patients and also to the increased urinary sodium clearance. Similarly, another study has also reported significantly higher serum phosphorous in Nigerian SCD children than in control.⁶ Contrary to our findings, some studies,^{36,37} reported lower serum phosphate in children with sickle disease which they linked it to their renal phosphate re-absorption as a result of increased parathyroid

hormone secretions. Their results could be due to the fact that; their subjects are SCD patients with renal disease.

Correlation study showed that there is a strong positive correlation between all the studied electrolytes in both crisis and steady

Table 4. Serum Antioxidant Vitamins Levels among the Age groups of the SCD Patients

Parameters	Age Groups				p-value
	0-10years	11-20years	21-30years	31years and above	
Vitamin A (µmol/L)	0.84±0.09	0.73± 0.09	0.68± 0.06	0.68± 0.77	0.070
Vitamin C (µmol/L)	21.88±1.27	21.08± 1.58	20.65± 1.04	19.10± 2.13	0.260
Vitamin E (µmol/L)	10.87±0.62	10.45±0.61	9.92± 0.30	9.90± 0.52	0.200

health conditions. This suggested that the electrolytes exhibited a closely parallel physiological changes with each other during acute crisis as well as in a steady health condition. This is in agreement with another study⁶ that reported a positive correlation between serum Phosphorous and Magnesium and also between serum Calcium and Magnesium in SCD patients.

This study also evaluated plasma levels of antioxidant vitamins A, C, and E, finding that these were significantly lower in SCD patients compared to those with sickle cell trait and healthy controls. This finding aligns with previous studies,^{39,40} which reported decreased antioxidant vitamins A, C and E levels in SCD patients. The depletion of these vitamins, particularly vitamins C and E, may be attributed to increased oxidative stress in SCD, which results in higher utilization and deficiency of these vitamins. Increased intake of synthetic or natural antioxidants could potentially help in mitigating the complications associated with SCD.

Conclusion

This study reveals significant variations in serum electrolyte levels among sickle cell disease (SCD) patients during vaso-occlusive crises, those in steady health, and healthy controls. The SCD patients in crisis exhibited lower calcium and magnesium levels but higher

phosphorus levels compared to the other groups, indicating disruptions in electrolyte homeostasis during acute episodes. The correlation analysis suggested complex relationships between electrolyte levels and disease severity. Additionally, the study found significantly reduced antioxidant vitamin levels in SCD patients, potentially contributing to oxidative stress and disease progression. These findings suggest that dietary antioxidants may help restore redox balance and reduce oxidative stress in SCD patients.

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Conflict of Interest

The authors declare that there are no conflicts of interest.

Copyright Statement

The authors affirm that:

- (i) This study has not been previously published and is not currently under consideration for publication elsewhere.
- (ii) All authors have approved this submission, and if the manuscript is accepted for publication, it will not be submitted for publication in any form or language without the written consent of the Editor-In-Chief.

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